Spam Message Classification Using Naive Bayes and Logistic Regression

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***Abstract*—Short Message Service (SMS) spam filtering remains challenging due to strict 160-character payload limits, informal abbreviations, and severe class imbalance. This paper benchmarks generative Multinomial Naive Bayes (MNB) and discriminative L_2-regularized Logistic Regression (LR) over sublinear TF-IDF unigram and bigram features on the UCI SMS Spam Collection ($N = 5574$). Evaluated on a stratified hold-out test set ($N = 1394$), LR achieves 0.9775 Average Precision, 94.1% F_1-score, and a 0.08% false positive rate, outperforming MNB (0.9732 AP, 93.3% F_1, 0.25% FPR) while maintaining better probability calibration.**

***Index Terms*—SMS spam filtering, Multinomial Naive Bayes, Logistic Regression, TF-IDF, class imbalance, Precision-Recall**

I. Introduction

Mobile text messaging is one of the most widely used communication channels worldwide, making Short Message Service (SMS) a prime target for unsolicited commercial advertisements, phishing lures, and financial fraud [1], [2]. Unlike email spam filtering—where classifiers can inspect routing headers, HTML markup, and hundreds of tokens per document [3]—content-based SMS spam filters operate under tight structural constraints. Standard SMS messages are capped at 140 bytes (160 seven-bit ASCII characters), resulting in very few words per message and extreme feature sparsity [1]. Moreover, mobile text is rife with phonetic slang, abbreviations, and deliberate character substitutions (such as "txt", "gr8", or "150p") that degrade standard lexical tokenizers [2].

From an operational standpoint, SMS spam filtering is governed by an asymmetric cost structure: a filter must maximize spam detection recall while keeping the false positive rate (FPR) as close to zero as possible [1], [3]. Blocking a legitimate personal or emergency message (*ham*) is far more disruptive to a user than letting an occasional advertisement reach the inbox [2]. This trade-off is compounded by the skewed class distribution of real-world messaging traffic, where spam typically accounts for less than 15% of volume [1], [4]. As shown by Davis and Goadrich [5], Receiver Operating Characteristic (ROC) curves can present an overly optimistic view on skewed datasets because the large number of true negatives dominates the FPR denominator. Rigorous evaluation therefore requires Precision-Recall (PR) curves alongside explicit threshold sensitivity analysis [5], [6].

In this paper, we present a reproducible empirical study comparing classical generative and discriminative linear classifiers—Multinomial Naive Bayes (MNB) and L_2-regularized Logistic Regression (LR)—on the public UCI SMS Spam Collection [1], [4]. Our main contributions are threefold:

1) We formulate and compare Multinomial Naive Bayes [3], [7] and L_2-regularized Logistic Regression [6], [8] over sublinear TF-IDF unigram and bigram representations [9].
2) We evaluate white-box feature interpretability by contrasting learned LR regression weights (β_j) with MNB log-likelihood ratios across the most informative spam tokens.
3) We analyze probability calibration and operational deployment by mapping precision, recall, and false positive rates across decision thresholds $\tau \in [0.05, 0.95]$.

II. Related Work

A. SMS and Email Spam Filtering

Statistical spam filtering was first studied extensively in electronic mail. Metsis et al. [3] evaluated five Naive Bayes variants across six Enron-Spam corpora, showing that Multinomial Naive Bayes and Flexible Bayes achieve strong cost-sensitive performance when false positives are heavily penalized. Extending these methods to mobile messaging, Almeida et al. [1] introduced the UCI SMS Spam Collection [4] and benchmarked 14 classifiers, including Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), C4.5 decision trees, and Naive Bayes variants. Because email filters degrade on short, noisy texts, Almeida et al. [2] later combined lexicographic normalization (using Lingo and English dictionaries) with semantic concept expansion from BabelNet to resolve out-of-vocabulary slang, polysemy, and synonymy in SMS and instant messaging spam.

B. Generative vs. Discriminative Text Categorization

McCallum and Nigam [7] clarified the distinction between the multi-variate Bernoulli event model (binary word presence) and the multinomial event model (word frequency counts), demonstrating that the multinomial formulation reduces classification error by 27% on average across text corpora. Despite its conditional independence assumption, Naive Bayes competes well with complex classifiers because zero-one classification accuracy remains optimal as long as the estimated

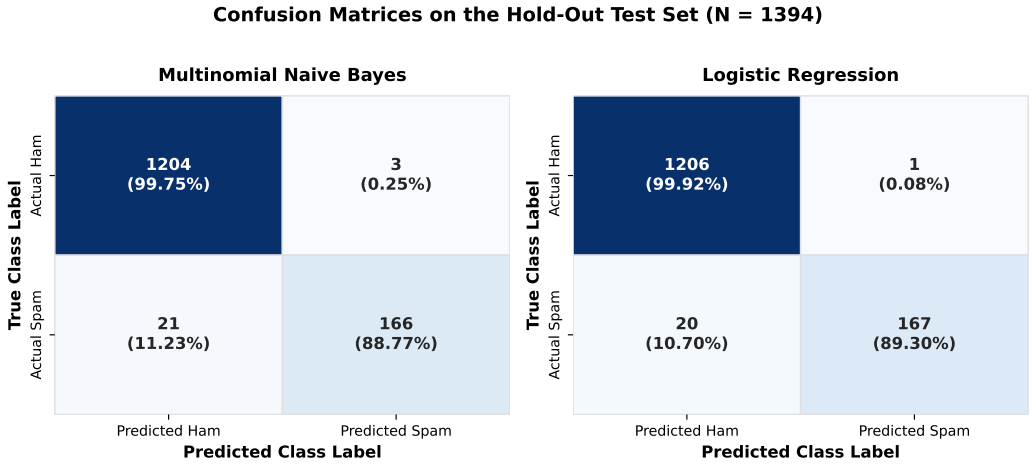


Fig. 1. Confusion matrices on the hold-out test set ($N = 1394$) at the default threshold $\tau = 0.50$. Logistic Regression yields only a single false positive (0.08% FPR) compared to 3 false positives (0.25% FPR) for Multinomial Naive Bayes.

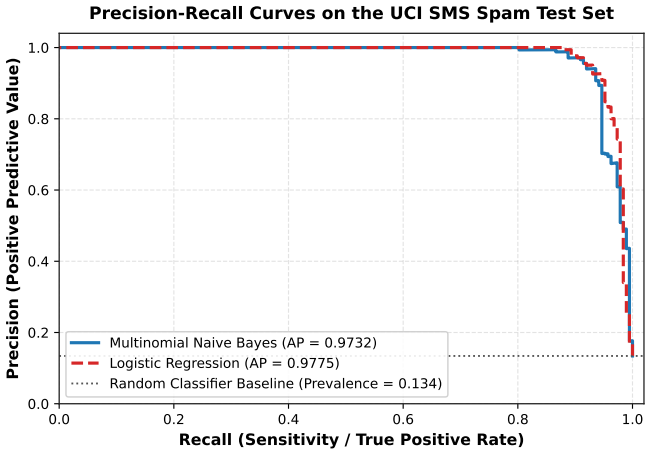


Fig. 2. Precision-Recall curves on the UCI SMS Spam hold-out test set ($N = 1394$). L_2-regularized Logistic Regression achieves a higher Average Precision (AP = 0.9775) than Multinomial Naive Bayes (AP = 0.9732).

posterior agrees with the true distribution on the argmax class, or when feature dependencies cancel out [10].

Comparing generative Naive Bayes and discriminative Logistic Regression, Ng and Jordan [8] proved that while discriminative learning achieves a lower asymptotic error floor, generative Naive Bayes converges toward its asymptotic error at an $\mathcal{O}(\log n)$ rate rather than $\mathcal{O}(n)$. Dreiseitl and Ohno-Machado [6] emphasized that Logistic Regression acts as a parametric white-box model whose coefficients directly measure feature importance, while weight decay (L_2 shrinkage) smooths the decision boundary to prevent overfitting. Furthermore, Zhang et al. [9] showed that combining TF-IDF weighting with multi-word sequences captures local context while retaining strong statistical discriminative power in high-dimensional spaces.

III. Data

We use the benchmark UCI SMS Spam Collection [1], [4], which comprises 5574 English SMS messages compiled from four sources: 425 spam messages from the UK Grumbletext forum, 3375 ham messages from the NUS SMS Corpus, 450 ham messages from Tagg's doctoral thesis, and 1324 messages (1002 ham, 322 spam) from the SMS Spam Corpus v.0.1 Big [1]. Overall, the dataset contains 4827 legitimate (*ham*, 86.60%) and 747 unsolicited (*spam*, 13.40%) messages. Spam messages average 23.48 tokens (17 543 tokens total), whereas ham messages average 13.18 tokens (63 632 tokens total) [1].

To preserve the 13.40% spam prior and obtain an unbiased generalization error estimate [6], we apply a stratified 75%/25% train–test split, yielding $N_{\text{train}} = 4180$ training messages (3620 ham, 560 spam) and a hold-out test set of $N_{\text{test}} = 1394$ messages (1207 ham, 187 spam). Following Almeida et al. [1], we omit stop-word removal and stemming so as not to erase informative triggers. Messages are lowercased and tokenized via a regular expression pattern that preserves alphanumeric strings, monetary rates ("150p", "150ppm"), numbers ("18", "500", "1000"), and domain fragments ("www", "uk", "com") while splitting on punctuation [1]. To capture both individual words and multi-word phrases [9], we extract unigrams and bigrams ($n \in \{1, 2\}$) with minimum document frequency df ≥ 2.

IV. Methods

A. TF-IDF Feature Representation

Each SMS message d is mapped into a vector $\mathbf{x} \in \mathbb{R}^V$ using sublinear Term Frequency–Inverse Document Frequency (TF-IDF) weighting [9]:

$$w_{t,d} = (1 + \log \text{tf}(t, d)) \cdot \left(\log \frac{1 + N_{\text{train}}}{1 + \text{df}(t)} + 1 \right), \qquad (1)$$

where $\text{tf}(t, d)$ is the frequency of term t in d and $\text{df}(t)$ is its training document frequency. Because spam messages are

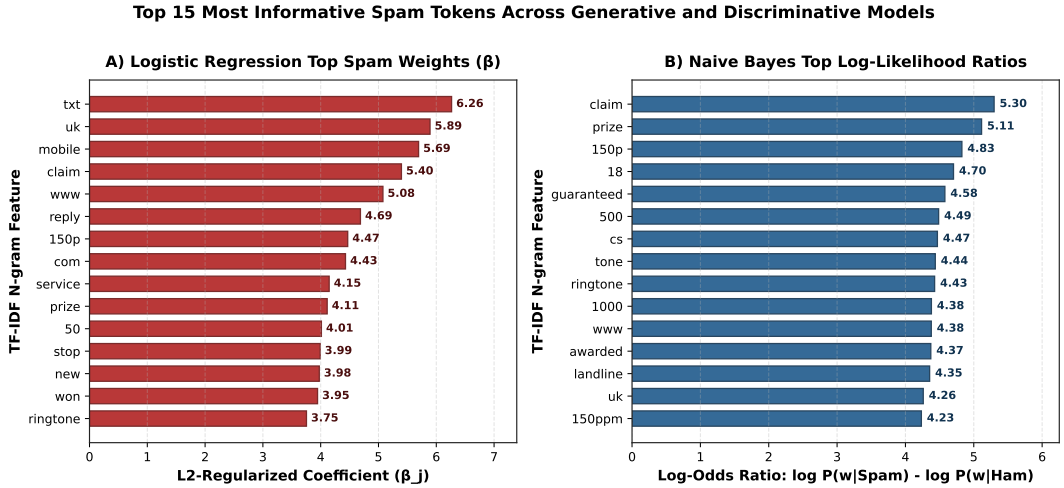


Fig. 3. Top 15 most informative spam tokens across (A) L_2-regularized Logistic Regression weights (β_j) and (B) Multinomial Naive Bayes log-likelihood ratios.

nearly twice as long as ham messages on average [1], vectors are L_2-normalized ($\tilde{\mathbf{x}} = \mathbf{x}/\|\mathbf{x}\|_2$) to eliminate length bias [3].

B. Multinomial Naive Bayes (MNB)

Under the multinomial event model [3], [7], the posterior log-odds between spam ($y = 1$) and ham ($y = 0$) is a linear function of $\tilde{\mathbf{x}}$ [8]:

$$\log \frac{P(y=1 \mid \tilde{\mathbf{x}})}{P(y=0 \mid \tilde{\mathbf{x}})} = \log \frac{\hat{\pi}_1}{\hat{\pi}_0} + \sum_{j=1}^{V} \tilde{x}_j \log \frac{\hat{\theta}_{j|1}}{\hat{\theta}_{j|0}}, \tag{2}$$

where $\hat{\pi}_c = N_c/N_{\text{train}}$ is the empirical class prior and $\hat{\theta}_{j|c}$ is estimated via Laplace smoothing ($\alpha > 0$) [3], [7]:

$$\hat{\theta}_{j|c} = \frac{\sum_{i:y^{(i)}=c} \tilde{x}_j^{(i)} + \alpha}{\sum_{k=1}^{V} \sum_{i:y^{(i)}=c} \tilde{x}_k^{(i)} + \alpha V}. \tag{3}$$

Token importance in MNB is quantified by the log-likelihood ratio $\text{LLR}(t_j) = \log \hat{\theta}_{j|1} - \log \hat{\theta}_{j|0}$.

C. Discriminative L_2-Regularized Logistic Regression

Logistic Regression directly models the posterior $P(y = 1 \mid \tilde{\mathbf{x}}) = \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}} + b)$ via the logistic sigmoid $\sigma(z) = (1+e^{-z})^{-1}$ [6], [8]. Parameters $(\boldsymbol{\beta}, b)$ are learned by minimizing the L_2-penalized binary cross-entropy loss [6]:

$$\mathcal{L}(\boldsymbol{\beta}, b) = -\sum_{i=1}^{N_{\text{train}}} \Big[y^{(i)} \log p_i + (1 - y^{(i)}) \log(1 - p_i) \Big] + \frac{\lambda}{2} \|\boldsymbol{\beta}\|_2^2, \tag{4}$$

where $p_i = \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}^{(i)} + b)$ and $\lambda > 0$ controls weight shrinkage, smoothing the decision boundary to prevent overfitting on correlated n-grams [6].

V. Results and Discussion

A. Precision-Recall and Confusion Matrix Evaluation

Given the skewed test distribution (1207 ham vs. 187 spam messages, 13.4% prevalence), we evaluate classifiers in Precision-Recall space [5]. Fig. 2 shows the PR curves for MNB and LR, and Table I summarizes all metrics. Both linear classifiers maintain nearly 100% precision up to 80% recall. Across the full recall range, L_2-regularized Logistic Regression achieves AP = 0.9775, outperforming Multinomial Naive Bayes (AP = 0.9732). Consistent with Ng and Jordan [8], at $N_{\text{train}} = 4180$, discriminative training reaches its lower asymptotic error regime, whereas MNB is slightly degraded by correlated unigrams and bigrams. Both models train in milliseconds (0.002 s for MNB and 0.015 s for LR).

TABLE I
Classification Performance on the Hold-Out Test Set ($N = 1394$)

Model	Prec. (%)	Rec. (%)	F_1 (%)	FPR (%)	AP
Multinomial NB	98.22	88.77	93.26	0.25	0.9732
Logistic Regression	**99.40**	**89.30**	**94.08**	**0.08**	**0.9775**

At the default threshold $\tau = 0.50$ (Fig. 1), MNB misclassifies 3 of 1207 ham messages (TN = 1204, FP = 3, FPR = 0.25%) and detects 166 of 187 spam messages (TP = 166, FN = 21, Recall = 88.77%), yielding 98.22% precision and 93.26% F_1. Logistic Regression cuts false positives to a single message (TN = 1206, FP = 1, FPR = 0.08%) while catching 167 spam messages (TP = 167, FN = 20, Recall = 89.30%), achieving 99.40% precision and 94.08% F_1. Both models improve substantially upon the unigram baselines in Almeida et al. [1] (where linear SVM caught 83.10% of spam at 0.18% FPR, and MNB caught 52.06%).

B. Feature Interpretability: Generative vs. Discriminative

Inspecting white-box model parameters [6] in Fig. 3 reveals how LR weights β_j (Panel A) and MNB log-odds ratios (Panel B) rank the top 15 spam tokens. Both models highlight commercial triggers such as `claim` ($\beta = 5.40$, LLR = 5.30), `prize` ($\beta = 4.11$, LLR = 5.11), `150p` ($\beta = 4.47$,

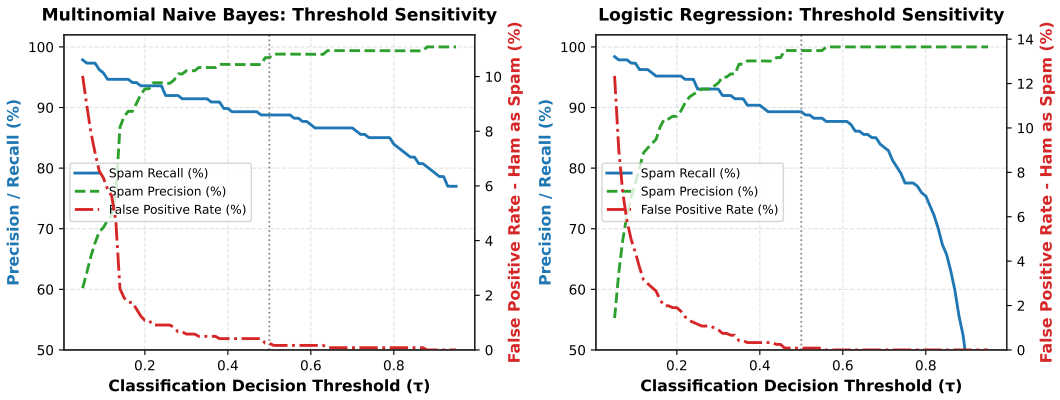


Fig. 4. Impact of classification decision threshold (τ) on Spam Precision, Recall, and False Positive Rate (FPR) for Multinomial Naive Bayes and Logistic Regression.

LLR $= 4.83$), `www` ($\beta = 5.08$, LLR $= 4.38$), `uk` ($\beta = 5.89$, LLR $= 4.26$), and `ringtone` ($\beta = 3.75$, LLR $= 4.43$).

However, because MNB estimates conditional probabilities independently [7], [10], its top log-odds ratios favor narrow, campaign-specific tokens that rarely occur in ham (`18` LLR $= 4.70$, `guaranteed` LLR $= 4.58$, `500` LLR $= 4.49$, `1000` LLR $= 4.38$, `150ppm` LLR $= 4.23$). Conversely, joint optimization under L_2 shrinkage [6] leads LR to assign its largest weights to broad call-to-action tokens (`txt` $\beta = 6.26$, `uk` $\beta = 5.89$, `mobile` $\beta = 5.69$, `reply` $\beta = 4.69$, `service` $\beta = 4.15$, `stop` $\beta = 3.99$), avoiding redundant double-counting.

C. Decision Threshold Sensitivity and Calibration

To assess probability calibration across operating points [3], [6], Fig. 4 plots precision, recall, and FPR across $\tau \in [0.05, 0.95]$. Below $\tau = 0.20$, both models achieve $> 95\%$ recall but suffer from excessive false positive rates (10%–13%). Above $\tau = 0.25$, FPR drops below 1%. Because MNB multiplies correlated token likelihoods, its posteriors are pushed toward 0 and 1 [3], [10], resulting in a gradual recall decay from 88.8% at $\tau = 0.50$ to 77.0% at $\tau = 0.95$. In contrast, LR exhibits a well-calibrated plateau across $\tau \in [0.45, 0.65]$, maintaining 99.4%–100% precision and $\approx 88\%$–89.3% recall at $\leq 0.08\%$ FPR, confirming $\tau = 0.50$ as a robust operating threshold.

VI. Conclusion

We replicated and compared Multinomial Naive Bayes and L_2-regularized Logistic Regression on the UCI SMS Spam Collection [1], [4]. Coupling sublinear TF-IDF unigram and bigram features with L_2-regularized Logistic Regression achieves 99.40% precision, 94.08% F_1-score, AP $= 0.9775$, and a 0.08% FPR in just 15 ms of training time, outperforming Multinomial Naive Bayes in both discriminative accuracy and probability calibration.

References

[1] T. A. Almeida, J. M. G. Hidalgo, and A. Yamakami, "Contributions to the study of sms spam filtering: new collection and results," in *Proceedings of the 11th ACM Symposium on Document Engineering*, ser. DocEng '11. New York, NY, USA: Association for Computing Machinery, 2011, p. 259–262. [Online]. Available: https://doi.org/10.1145/2034691.2034742

[2] T. A. Almeida, T. P. Silva, I. Santos, and J. M. Gómez Hidalgo, "Text normalization and semantic indexing to enhance instant messaging and sms spam filtering," *Knowledge-Based Systems*, vol. 108, pp. 25–32, 2016, new Avenues in Knowledge Bases for Natural Language Processing. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0950705116300909

[3] V. Metsis, I. Androutsopoulos, and G. Paliouras, "Spam filtering with naive bayes-which naive bayes?" in *CEAS*, vol. 17. Mountain View, CA, 2006, pp. 28–69.

[4] T. Almeida and J. Hidalgo, "SMS Spam Collection," UCI Machine Learning Repository, 2011, DOI: https://doi.org/10.24432/C5CC84.

[5] J. Davis and M. Goadrich, "The relationship between precision-recall and roc curves," in *Proceedings of the 23rd International Conference on Machine Learning*, ser. ICML '06. New York, NY, USA: Association for Computing Machinery, 2006, p. 233–240. [Online]. Available: https://doi.org/10.1145/1143844.1143874

[6] S. Dreiseitl and L. Ohno-Machado, "Logistic regression and artificial neural network classification models: a methodology review," *Journal of Biomedical Informatics*, vol. 35, no. 5, pp. 352–359, 2002. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1532046403000340

[7] A. McCallum, K. Nigam *et al.*, "A comparison of event models for naive bayes text classification," in *AAAI-98 workshop on learning for text categorization*, vol. 752, no. 1. Madison, WI, 1998, pp. 41–48.

[8] A. Ng and M. Jordan, "On discriminative vs. generative classifiers: A comparison of logistic regression and naive bayes," in *Advances in Neural Information Processing Systems*, T. Dietterich, S. Becker, and Z. Ghahramani, Eds., vol. 14. MIT Press, 2001. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2001/file/7b7a53e239400a13bd6be6c91c4f6c4e-Paper.pdf

[9] W. Zhang, T. Yoshida, and X. Tang, "A comparative study of tf* idf, lsi and multi-words for text classification," *Expert systems with applications*, vol. 38, no. 3, pp. 2758–2765, 2011.

[10] H. Zhang *et al.*, "The optimality of naive bayes," *Aa*, vol. 1, no. 2, p. 3, 2004.